Temporal Equilibrium MeanFlow: Bridging the Scale Gap for One-Step Generation

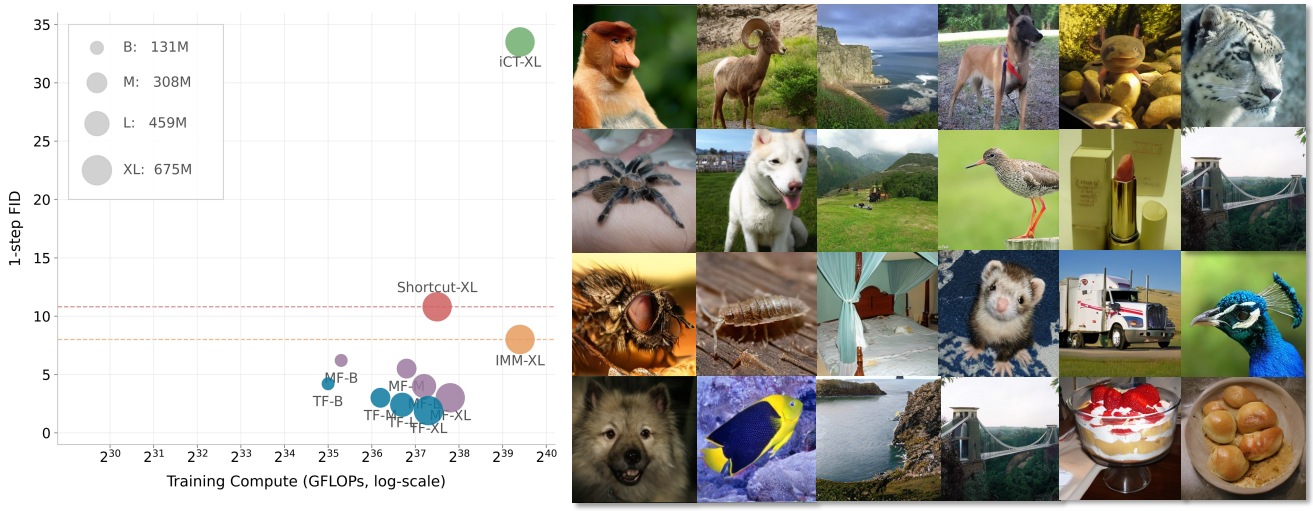

Figure 1. **One-Step Generation of 256×256 ImageNet Images from Scratch**. Our model delivers significantly superior generation quality compared to prior state-of-the-art one-step diffusion and flow methods. Specifically, iCT [30], Shortcut [8], MeanFlow [10], and our method all achieve generation with 1 network forward evaluation (1-NFE), whereas the 1-step result of IMM [41] requires 2-NFE for guidance. Detailed quantitative results are provided in Table 2, and all images presented herein are generated by our 1-NFE model.

Abstract

MeanFlow is a powerful few-step generative framework that can be trained from scratch, but its performance degrades significantly when the one-step loss uses a large portion of training data. This stems from a temporal scale imbalance: gradients from different stages of generation contribute unevenly, leading to unstable optimization—evident in blurry samples and high FID scores. The core issue is a conflict between two opposing forces: terms that amplify variance over long time spans and strong constraints needed near the start of generation, which a fixed sampling strategy cannot reconcile. To resolve this, we propose Temporal Equilibrium MeanFlow (TEMF), which balances these competing demands through two simple yet effective components: (1) a temporal equilibrium weighting function that equalizes gradient influence across all time scales, and (2) a dynamic boundary scheduler that gradually shifts training focus—from stabilizing early steps to refining the full trajectory as training progresses. Without changing the model architecture, TEMF retains true one-step generation with classifier-free guidance, achieving a state-of-the-art FID of 2.62 on ImageNet 256×256—achieving the best results among diffusion- and flow-based one-step methods.

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1. Introduction

Generative modeling seeks to learn transformations that map simple prior distributions to complex data distributions. Flow Matching [1, 17, 18] has emerged as a powerful framework for this task, learning time-dependent velocity fields that guide the transformation process through continuous-time dynamics. While conceptually related to diffusion models [14, 32], Flow Matching offers a more direct approach by directly regressing the transport velocity field, resulting in a concise objective formulation that has facilitated its widespread adoption in state-of-the-art generative systems [7, 21].

A critical limitation of both Flow Matching and diffusion models is their reliance on iterative sampling procedures, requiring multiple network evaluations (NFEs) to generate each image. This computational bottleneck has motivated extensive research into few-step—and particularly one-step—generative modeling. Among recent approaches, two families have shown particular promise. Consistency

Models [12, 19, 30, 31] learn to produce consistent outputs across different timepoints along the same trajectory, while MeanFlow [10] introduces average velocity fields that unify instantaneous and interval-based dynamics.

MeanFlow represents a significant methodological advance by deriving a well-defined identity that connects average and instantaneous velocities. This theoretical foundation enables stable one-step training from scratch, eliminating the need for distillation or carefully designed curriculum learning. However, practical implementation reveals a fundamental limitation: the model's performance degrades significantly when the proportion of trajectory samples (where $r \neq t$) exceeds an optimal threshold. This constraint prevents the model from fully leveraging the rich information contained in long-range transitions, ultimately limiting one-step generation quality.

Our analysis reveals that this limitation stems from a temporal scale imbalance: gradient contributions from different stages of the generation process are distributed unevenly, creating unstable training dynamics. Specifically, we identify a fundamental tension between two competing requirements: (1) Strong boundary constraints needed near the start of generation (where $r \approx t$), which provide training stability but limit trajectory flexibility. (2) Accurate trajectory modeling is necessary across longer time intervals (where $r \ll t$), which improves generation quality but introduces optimization instability. The fixed sampling strategy used in the MeanFlow cannot reconcile these competing demands, resulting in suboptimal performance and an unstable training process when attempting to increase the proportion of one-step trajectory samples during the training process.

To resolve this issue, we propose Temporal Equilibrium MeanFlow (TEMF), a novel framework that balances these competing requirements through two complementary mechanisms:(1) Temporal equilibrium weighting: a scale-aware weighting function that equalizes gradient influence across all time scales by counteracting the natural variance imbalance between short and long intervals. (2) Dynamic boundary scheduling: a mechanism that gradually shifts training focus from boundary stabilization to trajectory refinement as optimization progresses, recognizing that optimal boundary constraint strength varies throughout training. Unlike prior approaches that treat boundary conditions as static constraints, TEMF explicitly models their evolution across training stages. This insight leads to a two-stage training strategy that begins with strong boundary constraints for stability, then progressively relaxes these constraints to prioritize trajectory accuracy as optimization converges. Our contributions can be summarized as:

- We identify a fundamental limitation in one-step generation: temporal scale imbalance causes optimization instability when trajectory samples exceed an optimal threshold, preventing effective utilization of long-range information essential for one-step synthesis.
- We propose TEMF, a framework that combines temporal equilibrium weighting with boundary-guided optimization and introduces no additional sampling-time computational overhead, enabling stable training and more accurate one-step prediction while maintaining compatibility with standard MeanFlow architectures.
- Experimental results demonstrate that TEMF achieves the best performance with a FID of 2.62 on ImageNet 256×256, outperforming all existing diffusion- and flow-based one-step methods by a clear margin, and with just 2-NFE sampling, our FID further drops to 2.30.

2. Related Work

Diffusion and Flow Matching. Diffusion models [14, 29, 32, 33] corrupt clean data with progressively increasing Gaussian noise and train a time-conditioned network to reverse this process. The forward–reverse dynamics can be written as a stochastic differential equation (SDE) and, equivalently, as a deterministic probability-flow ordinary differential equation (ODE) [15, 33]. Flow Matching [1, 17, 18] adopts a closely related but more direct perspective: it learns the time-dependent *velocity field* that transports a simple prior distribution into the data distribution. Training proceeds by matching the model's velocity to a target field induced by stochastic interpolants or transport couplings. Viewed in this way, Flow Matching is naturally interpreted as a continuous-time normalizing flow [24].

Few-step Diffusion/Flow Models. Reducing the number of sampling steps (NFEs) is crucial for practical deployment. A common strategy distills a pre-trained many-step model into a few-step sampler [11, 26, 27] or uses score-based distillation [20, 37, 42]. Consistency Models [31] provide a teacher-free alternative: they enforce a functional *consistency* relation so that network outputs at different times map to the same endpoint, and numerous variants refine this idea and its training curricula [12, 19, 30, 31, 36]. Recent work further studies objects that depend on *pairs* of time instants. Boffi *et al*. [3] introduce a *Flow Map*—the path integral of the velocity between two times—and learn it via matching losses; compared with the average velocity used in our approach, the Flow Map represents total displacement over an interval. Shortcut Models [8] add cross-interval self-consistency on top of Flow Matching, while Inductive Moment Matching [41] enforces moment constraints for stochastic interpolants across different times, offering another route to two-time consistency. SoFlow and AlphaFlow [2, 40] focus on improving the computation efficiency and the coupled items within the loss, respectively.

3. Preliminary: MeanFlow

We begin by specifying the MeanFlow setup to establish notation and identify limitations. Given data distribution

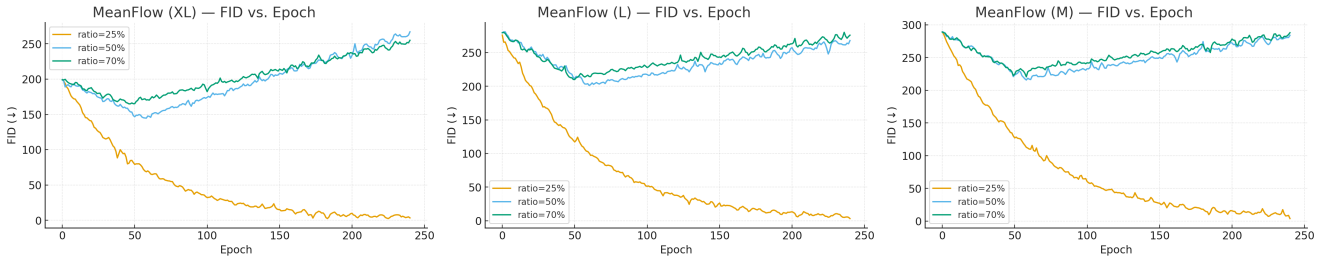

Figure 2. **FID-50K performance of MeanFlow for 1-NFE when adopting different models (DiT-XL/M/L) and $r \neq t$ sample ratios.** It can be observed that the performance is quite sensitive to the ratio. When using a much larger ratio, MeanFlow [10] suffers from a significant performance degradation.

$p(x_0)$ and prior distribution $p(x_1) = N(0, I)$, a general noising process is defined as $x_t = \alpha_t x_0 + \beta_t x_1$, where $t \in [0, 1]$, and α_t, β_t are continuously differentiable functions satisfying boundary conditions $\alpha_0 = 1, \alpha_1 = 0$, $\beta_0 = 0$, and $\beta_1 = 1$. The average velocity field in MeanFlow is defined as:

$$u(x_t, r, t) \triangleq \frac{1}{t-r} \int_r^t v(x_\tau, \tau) d\tau, \tag{1}$$

where $v(x_t, t)$ is the instantaneous velocity field. Through differential calculus, this leads to the original MeanFlow Identity:

$$u(x_t, r, t) = v(x_t, t) - (t-r) \frac{d}{dt} u(x_t, r, t), \tag{2}$$

with total derivative (treating r as constant)

$$\frac{d}{dt} u = v(x_t, t) \partial_x u + \partial_t u.$$

The MeanFlow training objective minimizes:

$$\begin{aligned} \mathcal{L}_{\text{MF}}(\theta) = \mathbb{E}_{x_t, r, t} \Big[\big\| u_\theta(x_t, r, t) - \text{sg}\big(v_t \\ - (t-r)\big(v_t \partial_x u_\theta + \partial_t u_\theta\big)\big) \big\|_2^2 \Big]. \end{aligned} \tag{3}$$

$v_t = \alpha_t' x_0 + \beta_t' x_1$ is the conditional velocity, and $sg(\cdot)$ denotes stop-gradient. A critical limitation emerges when examining the variance of the target term across different time scales. Define $\Delta = t - r$ as the time interval length. The target term's variance exhibits a strong growth for small Δ that can be well approximated by an $O(1/\Delta)$ dependence, creating a significant imbalance between short and long intervals. This imbalance manifests in training dynamics: when the proportion of $r \neq t$ samples exceeds the best proportion given in [10], the model's performance deteriorates due to unstable gradient updates and insufficient boundary constraint enforcement, as shown in Fig. 2.

4. Temporal Equilibrium MeanFlow

4.1. Temporal Scale Analysis

We begin by analyzing the fundamental cause of MeanFlow's limitations. The gradient contribution of samples with a time interval Δ is crucial for training stability. For clarity, we consider the gradient with respect to model parameters θ:

$$\nabla_\theta \mathcal{L}_{\text{MF}} = 2\mathbb{E}[(u_\theta(x_t, r, t) - u_{\text{tgt}}) \cdot \nabla_\theta u_\theta(x_t, r, t)], \tag{4}$$

where $u_{\text{tgt}} = v_t - \Delta(v_t \partial_x u_\theta + \partial_t u_\theta)$. The variance of this gradient has two components: (1) Boundary constraint variance: When $\Delta \to 0$, the term $u_\theta(x_t, r, t) - v_t$ dominates, which is empirically observed to grow roughly as $\mathcal{O}(1/\Delta)$ due to path curvature. (2) JVP amplification variance: When Δ is large, the term $\Delta(v_t \partial_x u_\theta + \partial_t u_\theta)$ dominates, with variance proportional to $\mathcal{O}(\Delta^2)$. This motivates the following theorem, which formalizes this intuition under mild regularity assumptions:

Theorem 1 (Temporal Variance Imbalance): The variance of the gradient contribution follows:

$$\text{Var}[\nabla_\theta \mathcal{L}_{\text{MF}}] \leq \frac{C_1}{\Delta} + C_2 \Delta^2 + \mathcal{O}(\Delta^3). \tag{5}$$

where $C_1, C_2 > 0$ are constants determined by the data distribution and model architecture, and it is built on mild regularity/curvature assumptions.

Proof 1. From the MeanFlow identity, we have: $\text{Var}[u_{\text{tgt}}] = \text{Var}[v_t - \Delta(v_t \partial_x u_\theta + \partial_t u_\theta)]$. The conditional velocity $v_t = \alpha_t' x_0 + \beta_t' x_1$ has variance:

$$\text{Var}[v_t] = (\alpha_t')^2 \, \text{Var}[x_0] + (\beta_t')^2 \, \text{Var}[x_1] \leq c_v, \tag{6}$$

where $c_v > 0$ is constant. The JVP term $(v_t \partial_x u_\theta + \partial_t u_\theta)$ has variance:

$$\text{Var}[v_t \partial_x u_\theta + \partial_t u_\theta] = c_j \tag{7}$$

where $c_j > 0$ depends on trajectory curvature. Combining these, we obtain the following decomposition:

$$\mathrm{Var}[u_{\text{tgt}}] = c_v + 2\Delta \cdot \mathrm{Cov}(v_t, v_t\partial_x u_\theta + \partial_t u_\theta) + \Delta^2 c_j. \quad (8)$$

The first two terms remain bounded as $\Delta \to 0$, while the $\Delta^2 c_j$ term dominates as Δ grows. In addition, empirical measurements of the target variance across time intervals (see Fig. 2) show a curvature-induced growth for very small Δ that can be well approximated by an extra term of order C_1/Δ. We therefore adopt the following heuristic temporal variance model:

$$\mathrm{Var}[u_{\text{tgt}}] \;\le\; \frac{C_1}{\Delta} + C_2\Delta^2 + \mathcal{O}(\Delta^3), \quad (9)$$

for some constants $C_1, C_2 > 0$.

Furthermore, the per-sample gradient can be written as

$$\nabla_\theta \mathcal{L}_{\text{MF}} = 2\,(u_\theta(x_t, r, t) - u_{\text{tgt}}) \cdot \nabla_\theta u_\theta(x_t, r, t), \quad (10)$$

and under mild regularity/curvature assumptions, we have a uniform bound $\|\nabla_\theta u_\theta(x_t, r, t)\|_2 \le C_J$. Standard variance inequalities then imply

$$\mathrm{Var}[\nabla_\theta \mathcal{L}_{\text{MF}}] \;\le\; C\,\mathrm{Var}[u_{\text{tgt}}] + \mathcal{O}(1), \quad (11)$$

for some constant $C > 0$ depending on C_J and the moments of u_θ. Substituting the temporal variance model above yields

$$\mathrm{Var}[\nabla_\theta \mathcal{L}_{\text{MF}}] \;\le\; \frac{C_1'}{\Delta} + C_2'\Delta^2 + \mathcal{O}(\Delta^3), \quad (12)$$

for suitable constants $C_1', C_2' > 0$, which completes the proof. □

This theorem reveals a fundamental tension in MeanFlow training: short intervals (i.e., Δ<0.1) require strong boundary constraints for stability, while long intervals (i.e., $\Delta > 0.5$) need precise trajectory modeling for accuracy. The fixed sampling strategy in the original MeanFlow fails to balance these competing requirements.

4.2. Temporal Equilibrium Weighting

To address the variance imbalance, we introduce a scale-aware weighting function that equalizes gradient contributions across all time scales. Our goal is to find $w(\Delta)$ such that:

$$\mathrm{Var}[\nabla_\theta L_{\text{TEMF}} \mid \Delta] = \text{constant} \quad (13)$$

Since, for a fixed interval length Δ, the weighted loss satisfies $\nabla_\theta L_{\text{TEMF}} \approx w(\Delta)\,\nabla_\theta L_{\text{MF}}$, Theorem 1 implies that we require:

$$w(\Delta)^2\left(\frac{C_1}{\Delta} + C_2\Delta^2\right) = \text{constant} \quad (14)$$

The solution to this equation is:

$$w(\Delta) = \frac{1}{\sqrt{\frac{C_1}{\Delta} + C_2\Delta^2}} = \frac{\sqrt{\Delta}}{\sqrt{C_1 + C_2\Delta^3}} \quad (15)$$

This is implemented via the following bounded parametric approximation:

$$w(\Delta) = \frac{1}{(1 + \lambda_1\Delta^{\beta_1}) \cdot \sqrt{1 + \lambda_2\Delta^2}}, \quad (16)$$

where $\lambda_1, \lambda_2, \beta_1 > 0$ are hyper-parameters. This function has two components: (1) The first term $1/\big(1 + \lambda_1\Delta^{\beta_1}\big)$ balances the boundary constraint strength. (2) The second term $1/\sqrt{1 + \lambda_2\Delta^2}$ counteracts JVP amplification for large Δ. Thus, our weighted loss function becomes:

$$\mathcal{L}_{\text{TEMF}}(\theta) = \mathbb{E}_{x_t, r, t}\Big[w(t - r) \cdot \|u_\theta(x_t, r, t) - \mathrm{sg}(u_{\text{tgt}})\|_2^2\Big] \quad (17)$$

Theorem 2 (Gradient Variance Bound): With the temporal equilibrium weighting function, for any $\Delta \in [\Delta_{\min}, 1]$ (with fixed $\Delta_{\min} > 0$) the gradient variance is uniformly bounded:

$$\sup_{\Delta \in [\Delta_{\min}, 1]} \mathrm{Var}[\nabla_\theta \mathcal{L}_{\text{TEMF}} \mid \Delta] < \infty. \quad (18)$$

A practical truncation $\Delta \ge \Delta_{\min} > 0$ is assumed in sampling.

Proof 2. Using the parametric weighting $w(\Delta)$, which is chosen as a bounded approximation to the ideal variance-equalizing solution, we obtain

$$\begin{aligned}\mathrm{Var}[\nabla_\theta L_{\text{TEMF}} \mid \Delta] \;&\le\; w(\Delta)^2\left(\frac{C_1}{\Delta} + C_2\Delta^2\right) \\ &= \frac{1}{(1 + \lambda_1\Delta^{\beta_1})^2(1 + \lambda_2\Delta^2)}\left(\frac{C_1}{\Delta} + C_2\Delta^2\right) \le K.\end{aligned} \quad (19)$$

Where $K = \sup_{\Delta \in [\Delta_{\min}, 1]} \frac{C_1/\Delta + C_2\Delta^2}{(1+\lambda_1\Delta^{\beta_1})^2(1+\lambda_2\Delta^2)} < \infty$ is a constant depending on $(C_1, C_2, \lambda_1, \lambda_2, \beta_1, \Delta_{\min})$.

4.3. Dynamic Boundary Scheduling

To further address the tension between boundary constraints and trajectory modeling, we introduce a dynamic boundary scheduling mechanism. Define $\tau \in [0, 1]$ as the normalized training progress, where $\tau = 0$ represents initialization and $\tau = 1$ represents convergence.

Definition 1 (Boundary Constraint Strength). The boundary constraint strength at training progress τ is:

$$\Gamma(\tau) = \mathbb{E}_{x_t, t}\Big[\|u_\theta(x_t, t, t) - v_t\|_2^2\Big], \quad (20)$$

where $v_t = \alpha'_t x_0 + \beta'_t x_1$ is the conditional velocity. This measures how well the model satisfies the boundary condition $u(x_t, t, t) = v_t$.

Theorem 3 (Heuristic Boundary Trajectory). A heuristic analysis suggests that the boundary constraint strength approximately follows an exponential decay trajectory:

$$\Gamma^*(\tau) = \Gamma_0 e^{-\lambda\tau}, \tag{21}$$

where $\Gamma_0 > 0$ is the initial constraint strength and $\lambda > 0$ is the decay rate.

Proof 3. The one-step generation error can be decomposed as:

$$E_{\text{1step}} = E_{\text{bias}}(\Gamma) + E_{\text{var}}(p), \tag{22}$$

where $E_{\text{bias}}(\Gamma)$ is the bias error decreasing with stronger boundary constraints, and $E_{\text{var}}(p)$ is the variance error decreasing with larger $r \neq t$ ratio p. The optimal trade-off is found by solving:

$$\frac{d}{d\tau}(E_{\text{bias}}(\Gamma(\tau)) + E_{\text{var}}(p(\tau))) = 0, \tag{23}$$

Under the simplified approximation $E_{\text{bias}}(\Gamma) = a\Gamma^2$ and $E_{\text{var}}(p) = b/(1-p)$ used for intuition rather than a precise characterization of the training dynamics, this leads to the differential equation:

$$\frac{d\Gamma}{d\tau} = -\lambda\Gamma, \tag{24}$$

whose solution is $\Gamma(\tau) = \Gamma_0 e^{-\lambda\tau}$. To implement this desired trajectory, we define the scheduling function for the $r \neq t$ sampling ratio:

$$p(\tau) = p_{\min} + (p_{\max} - p_{\min}) \cdot \sigma(\kappa(\tau - \tau_0)), \tag{25}$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the sigmoid function, $p_{\min} = 0.1, p_{\max} = 0.9, \kappa = 8$ controls the transition sharpness, and $\tau_0 = 0.6$ is the transition point. This implements a two-stage training strategy: (1) Boundary stabilization phase ($\tau < \tau_0$): Focus on learning the boundary condition $u(x_t, t, t) = v(x_t, t)$ with predominantly $r = t$ samples. (2) Trajectory optimization phase ($\tau \geq \tau_0$): Emphasize learning accurate trajectories with predominantly $r \neq t$ samples. We provide the pseudo code of the complete TEMF training algorithm in the Appendix.

5. Classifier-free Guidance Integration

TEMF naturally integrates Classifier-Free Guidance (CFG) by training the model on the guided velocity field directly. For class label c and guidance strength ω, the guided marginal velocity field is:

$$v^{\text{cfg}}(x_t, t|c) = \omega v(x_t, t|c) + (1-\omega) v(x_t, t), \tag{26}$$

where $v(x_t, t \mid c)$ is the conditional marginal velocity and $v(x_t, t)$ is the unconditional marginal velocity. During training, we randomly drop the class label with probability 0.1, allowing the model to learn both conditional and unconditional generation. For samples with class labels, the target velocity becomes:

$$v_t^{\text{cfg}} = \omega(\alpha'_t x_0 + \beta'_t x_1) + (1-\omega) v_{\text{uncond}}, \tag{27}$$

where v_{uncond} is the model's unconditional velocity prediction. To stabilize training with large ω, following [10], we introduce a velocity mixing mechanism:

$$v^{\text{mix}} = m \cdot v_t^{\text{cfg}} + (1-m) \cdot v^{\text{pred}}, \tag{28}$$

where v^{pred} is the model's prediction of the guided velocity field, and $m \in (0, 1]$ is a mixing ratio that decreases with training progress:

$$m(\tau) = 0.3 \cdot (1-\tau) + 0.1\tau. \tag{29}$$

This mixing reduces the variance of the training target while maintaining guidance effectiveness. With this integration, TEMF supports one-step generation with CFG:

$$x_0 = x_1 - u_\theta(x_1, 0, 1). \tag{30}$$

Without additional sampling-time computational overhead, we can maintain the one-step generation property while achieving higher sample quality.

6. Experiments

6.1. Experimental Setting

Our primary experiments are carried out on the ImageNet 256×256 dataset [5]. Following recent trends in generative modeling, we work in the latent space of the same pre-trained VAE used in [10]. This VAE encodes each 256×256 image into a compact 32×32×4 latent tensor. To quantify sample quality, we use the Fréchet Inception Distance (FID) [13], evaluated on 50,000 generated images. For efficiency analysis, we track the number of network function evaluations (NFE), with special attention to the one-step generation regime (1-NFE). Notably, all TEMF models are trained from scratch—no pre-training, distillation, or external initialization is used, and baseline results are taken from their original papers unless otherwise specified. In our ablation study, we use the DiT-B/4 architecture [6] as developed in [23], trained for 240 epochs.

6.2. Ablation Study

Temporal equilibrium weighting function. Tab. 1 (a) demonstrates the importance of balancing gradient contributions across different time scales. The baseline with no weighting (6.52 FID) suffers from temporal variance imbalance, while using only the JVP term (5.13 FID) or only the

Table 1. **Ablation studies on temporal equilibrium components and hyperparameters.** All experiments are conducted on ImageNet 256×256 with DiT-B/4 architecture (131M parameters), trained for 400K iterations from scratch.

Temporal equilibrium weighting	FID-50K (1-NFE)
No weighting	6.52
Only JVP term	5.13
Only boundary term	4.86
Full weighting	**4.31**

(a) Temporal equilibrium weighting function

Boundary scheduling	FID-50K (1-NFE)
Static (25%)	5.72
Linear	4.86
Cosine	4.53
Exponential	**4.31**

(b) Dynamic boundary scheduling

Velocity mixing mechanism	FID-50K (1-NFE)
No mixing ($m = 1.0$)	5.24
Fixed mixing ($m = 0.25$)	4.58
Dynamic mixing	**4.31**

(c) Velocity mixing for CFG integration

Component ablation	FID-50K (1-NFE)
Baseline MeanFlow	6.17
+ Temporal weighting	5.26
+ Dynamic boundary	4.93
+ CFG integration	4.61
Full TEMF	**4.31**

(d) Incremental component ablation

$r \neq t$ ratio	FID-50K (1-NFE)
0.0	5.95
0.25	4.87
0.50	**4.31**
0.75	4.45
1.0	4.68

(e) $r \neq t$ sampling ratio sensitivity

λ_1 (boundary term)	FID-50K (1-NFE)
0.0	4.75
0.3	4.42
0.5	**4.31**
0.7	4.48
1.0	4.63

(f) λ_1 sensitivity for boundary constraint

λ_2 (JVP suppression)	FID-50K (1-NFE)
0.5	4.58
1.0	**4.31**
1.5	4.47
2.0	4.62

(g) λ_2 sensitivity for JVP amplification

β_1 (decay rate)	FID-50K (1-NFE)
0.5	4.43
1.0	**4.31**
1.5	4.38
2.0	4.50

(h) β_1 sensitivity for decay rate control

velocity mix ratio m	FID-50K (1-NFE), w=3.0
0.1	4.49
0.3	**4.31**
0.5	4.42
0.7	4.67

(i) CFG velocity mixing parameter sensitivity

boundary term (4.86 FID) provides partial improvements. The complete weighting function combining both components achieves the best performance (4.31 FID), validating our theoretical analysis that different time scales require different treatment to stabilize training.

Dynamic boundary scheduling. Tab. 1 (b) examines the impact of different scheduling strategies for the r=t sampling ratio. The static 25% ratio used in MeanFlow (5.72 FID) provides reasonable stability but limits trajectory modeling. While linear (4.86 FID) and cosine (4.53 FID) schedules show improvements, the exponential schedule (4.31 FID) best implements our two-stage training strategy, providing strong boundary constraints early and gradually emphasizing trajectory modeling later. This is consistent with Theorem 3's heuristic prediction of an exponential decay trajectory for boundary constraint strength.

Velocity mixing mechanism for CFG. Tab. 1 (c) reveals how training stability can be maintained while incorporating guidance. Without mixing (m =1.0), the high variance of the CFG velocity field leads to poor performance (5.24 FID). Fixed mixing (m =0.25) improves stability (4.58 FID), but our dynamic mixing strategy that reduces m as training progresses achieves the best results (4.31 FID), demonstrating that variance reduction is particularly important in early training stages.

Incremental component ablation Tab. 1 (d) validates the complementary nature of our proposed components. Starting from the MeanFlow baseline (6.17 FID), adding temporal weighting reduces FID to 5.26, further incorporating dynamic boundary scheduling lowers it to 4.93, and adding CFG integration brings it to 4.61. The full TEMF framework, combining all these components, achieves 4.31 FID.

Parameter sensitivity. Tab. 1 (e) confirms our hypothesis that a balanced approach between boundary constraints and trajectory modeling is essential, with the optimal balance achieved at a 50% ratio when using our temporal equilibrium framework. Tab. 1 (g) investigates the impact of the λ_2 hyperparameter, which controls how aggressively the temporal equilibrium weighting function suppresses Jacobian-Vector Product amplification for large time intervals. The results show a clear optimal value at λ_2=1.0. Lower values provide insufficient suppression of JVP variance, leading to

Algorithm 1 Temporal Equilibrium MeanFlow Training

Input: model $u_\theta(x_t, r, t)$; weighting parameters $\lambda_1, \lambda_2, \beta_1$; scheduling parameters $p_{\min}, p_{\max}, \kappa, \tau_0$.

Procedure:

for training step $k = 1$ to K **do**
  $\tau \leftarrow k/K$
  $p \leftarrow p_{\min} + (p_{\max} - p_{\min}) \cdot \sigma\big(\kappa(\tau - \tau_0)\big)$
  Sample $g \sim \mathcal{N}(-0.4, 1.0)$ and set $t \leftarrow \sigma(g)$ {sample $t \in (0,1)$ by sigmoid of a Gaussian}
  Sample $r \sim \mathrm{Uniform}(0,1)$
  if $r > t$ **then**
    swap r and t {ensure $t > r$}
  end if
  With probability p, sample $\epsilon_r \sim \mathcal{N}(0, 10^{-3})$ and set $r \leftarrow r + \epsilon_r$ {perturb r so that $r \neq t$}
  $r \leftarrow \mathrm{clip}(r, 0, 1)$
  if $r > t$ **then**
    swap r and t {re-enforce $t > r$ after perturbation}
  end if
  Sample $x_0 \sim p_{\text{data}}$, $\epsilon \sim \mathcal{N}(0, I)$
  $x_t \leftarrow \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$
  $v_t \leftarrow \alpha_t' x_0 + \beta_t' x_1$
  $u_{\text{pred}} \leftarrow u_\theta(x_t, r, t)$
  Compute JVP: $\mathrm{d}u/\mathrm{d}t \leftarrow v_t \cdot \partial_x u_{\text{pred}} + \partial_t u_{\text{pred}}$
  $u_{\text{tgt}} \leftarrow v_t - (t - r)\, \mathrm{d}u/\mathrm{d}t$
  $\Delta \leftarrow t - r$
  $w \leftarrow \dfrac{1}{\big(1 + \lambda_1 \Delta^{\beta_1}\big)\sqrt{1 + \lambda_2 \Delta^2}}$
  $\mathcal{L} \leftarrow w \cdot \big\| u_{\text{pred}} - \mathrm{sg}(u_{\text{tgt}}) \big\|_2^2$
  Update θ via gradient descent
end for

Output: trained model u_θ.

training instability, while higher values over-constrain the model and prevent it from learning accurate trajectory information. Tab. 1 (h) examines how the β_1 parameter affects the performance of our temporal equilibrium weighting function. The β_1 parameter governs how rapidly the weighting function transitions between boundary-constrained and trajectory-focused regimes. Tab. 1 (i) analyzes how the velocity mixing ratio m affects generation quality when classifier-free guidance (CFG) is applied during training. The optimal balance is achieved at m=0.3, where the model benefits from guidance while maintaining training stability.

6.3. Comparisons with Prior Works

ImageNet 256×256 comparisons. We evaluate our method against state-of-the-art one-step models for class-conditional ImageNet-256×256 generation in Tab. 2. Within the few-step consistency model family, our approach achieves a new best FID of 2.62 with a single network forward pass. These gains narrow the quality gap between one-step and many-step generative models. Moreover, with just 2-NFE sampling (i.e., one generation step using classifier-free guidance), our FID further drops to 2.30, matching or surpassing many leading many-step models such as DiT-XL, SiT-XL, all of which require 250×2 NFEs under the same backbone. This strongly supports the claim that few-step models can rival many-step counterparts in generation quality. Fig. 3 shows that 1-NFE FID-50K steadily improves as model capacity scales from base to extra-large, consistent with trends in DiTs. Finally, qualitative samples from smaller backbones in Fig. 4 also demonstrate high fi-

Method	Epochs	Params	NFE	FID↓
Generative Adversarial Networks				
BigGAN [4]	-	112M	1	6.95
StyleGAN-XL [28]	-	166M	1	**2.30**
Masked and Autoregressive Models				
MagViT-v2 [38]	1080	307M	64	1.78
LlamaGen-XL [34]	300	775M	576	2.62
VAR [35]	350	2.0B	10	1.80
MAR [16]	800	943M	64	**1.55**
RandAR-XL [22]	300	775M	256	2.22
Multi-step Diffusion Models				
LDM-4-G [25]	170	395M	250 × 2	3.60
MDTv2 [9]	700	676M	250 × 2	1.63
DiT-XL [23]	1400	675M	250 × 2	2.27
SiT-XL [21]	1400	675M	250 × 2	2.06
SiT-REPA-XL [39]	800	675M	250 × 2	**1.42**
Few-step Consistency Models				
iCT-XL† [30]	-	675M	1 × 2	20.30
Shortcut-XL [8]	250	675M	1 × 2	10.60
IMM-XL [41]	3840	675M	1 × 2	7.77
MeanFlow-B [10]	240	131M	1	6.17
MeanFlow-M [10]	240	308M	1	5.01
MeanFlow-L [10]	240	459M	1	3.84
MeanFlow-XL [10]	240	675M	1	3.43
SoFlow-B [2]	240	131M	1	5.06
SoFlow-M [2]	240	308M	1	4.20
SoFlow-L [2]	240	459M	1	3.72
SoFlow-XL [2]	240	675M	1	**3.35**
Ours-B	240	131M	1	4.31
Ours-M	240	308M	1	3.82
Ours-L	240	459M	1	3.26
Ours-XL	240	675M	1	2.81
Ours-XL+	240+60	675M	1	**2.62**
Ours-XL+	240+60	675M	2	**2.30**

Table 2. FID-50K results for class-conditional generation on ImageNet-256×256. ×2 denotes an NFE of 2 per sampling step incurred by CFG. † Results as reported in [10].

method	precond	NFE	FID
iCT [30]	EDM	1	2.83
ECT [12]	EDM	1	3.60
sCT [19]	EDM	1	2.97
IMM [41]	EDM	1	3.20
MeanFlow [10]	none	1	2.92
SoFlow [2]	none	1	2.92
Ours	none	1	**2.81**

Table 3. **Unconditional CIFAR-10.**

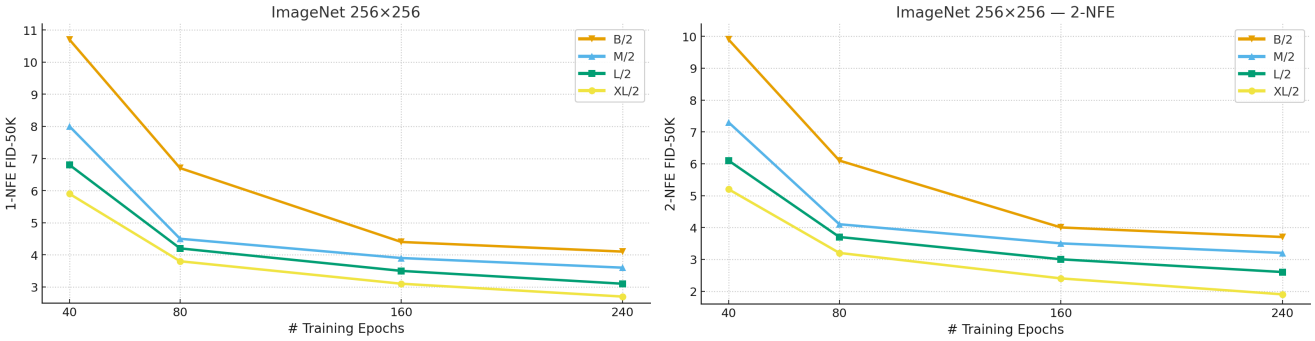

Figure 3. **FID-50K performance of our models for 1-NFE/2-NFE.** The figure presents the FID-50K scores of our models across different architectural scales, all trained from scratch on ImageNet at 256×256 resolution. Classifier-free guidance (CFG) is incorporated during training, and evaluation is performed using both 1-NFE and 2-NFE sampling protocols. The results exhibit the positive correlation between model size and generation quality in our framework.

Figure 4. **1-NFE Generation Results.** We show curated examples of class-conditional generation on ImageNet 256×256 using our 1-NFE model. It can be observed that our method can generate high-quality and diverse samples.

delity and diversity, confirming that compelling results do not require extremely large networks.

CIFAR-10 evaluation. We also evaluate unconditional image generation on CIFAR-10 at 32×32 resolution using FID-50K with a single network forward pass. As shown in Tab. 3, all methods use the same 55M-parameter U-Net architecture operating directly in pixel space; however, while prior approaches rely on EDM-style preconditioning, our method uses none. Despite this simpler setup, we achieve a new state-of-the-art FID. This result reinforces the consistent superiority of our approach across both low- and high-resolution generation benchmarks.

7. Conclusion

In this work, we present Temporal Equilibrium MeanFlow (TEMF), a framework for one-step generation that addresses the fundamental limitation of temporal scale imbalance in MeanFlow. We identified that gradient contributions vary dramatically across different time intervals, creating optimization instability when the proportion of non-boundary samples increases. TEMF introduces two key innovations to address this issue: (1) a temporal equilibrium weighting, (2) a dynamic boundary scheduling mechanism. These components work synergistically to stabilize training while maximizing trajectory information utilization. Experiments demonstrate consistent improvements over the baseline, achieving a FID of 2.62.

Limitations. While our TEMF demonstrates superior one-step generation performance, it has notable limitations. The framework introduces a small number of hyperparameters that require tuning across different model architectures. Additionally, the method still yields lower absolute image quality compared to multi-step diffusion models when computational resources permit additional function evaluations.

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